Josh Steckel

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EDUCATION

University of Delaware *Expected 2029*
Bachelor of Engineering (B.E.) – Electrical Engineering | GPA: 3.99 | Dean's List

SKILLS

Core Skills: PCB Design, RF and Digital Signal Processing, FPGA and Digital Design, Embedded Systems, Systems Programming
Hardware & Test Equipment: Oscilloscope, Spectrum Analyzer, Vector Network Analyzer (VNA), Multimeter
Programming Languages: C, C++, Python, MATLAB, VHDL
Software & Tools: KiCad, Vivado, GNU Radio, SolidWorks

EXPERIENCE

Ground Station Team Lead *05/2026 – Present*
Delaware Atmospheric Plasma Probe Experiment (DAPPEr)

- Led a 5-member sub team designing and upgrading ground station support equipment for a student-built LEO CubeSat mission
- Developed and tested channel models in MATLAB to determine optimal constant and adaptive downlink data rates for UHF spacecraft communications link
- Evaluated mission link budget and communications system performance for NASA's CubeSat Launch Initiative Pre-Procurement Readiness Review (PPRR) and FCC licensing
- Programmed command-handling software in Python to automate satellite pass scheduling and data acquisition using PLC, SDR, and rotor controller interfaces

Crew Member *05/2024 – 08/2024*
AMC Theatres, Inc., Allentown, PA

- Delivered guest services in a high-volume, fast-paced environment, operating concessions and maintaining a clean, organized, and safety-conscious space while collaborating with a large team

PROJECTS

Altoids Tin Meshtastic Node | *PCB Design, ESP32, LoRa, RF Design*

- Designed and assembled a 2-layer impedance-controlled RF PCB integrating an ESP32 MCU, UHF LoRa module, GPS module, user interface, and TFT LCD display
- Built and tested a LiPo battery-powered Meshtastic LoRa node housed in an Altoids tin, enabling decentralized, off-the-grid text communications and validating RF performance with lab test equipment

University Transit System Departure Board | *Embedded Systems, ESP32, FreeRTOS, Python, AWS*

- Designed a compact OLED arrival/departure board compatible with any ETA Transit-powered system, built on an automated headless web scraper (Selenium, Docker) hosted on AWS Lambda
- Implemented multi-core RTOS firmware on a low-power ESP32 client fetching pre-processed JSON data over secure HTTPS, cutting load time by 75% while running a horizontally scrolling marquee display

Fault Injection on PQC Algorithm Research | *Hardware Security, ChipWhisperer, ARM, Post-Quantum Cryptography*

- Contributing to VIP project team using ChipWhisperer boards to introduce instruction skips to ARM microcontrollers running Post-Quantum Cryptography (PQC) encryption algorithms (i.e. Kyber)
- Analyzing responses to tune glitch parameters to optimize fault injection windows

Power Side-Channel Attacks Research | *Machine Learning, CNNs, Signal Processing, Python*

- Trained and tested CNN models on 60+ hours of VR power consumption data to classify device activity from STFT frequency-spectrum images, achieving 97% classification accuracy and strengthening applied signal processing and data analysis skills

J-Pole Antenna Design and Installation | *Antenna Design, RF Engineering, Impedance Matching*

- Applied antenna theory to design and tune a 145 MHz VHF J-pole antenna, replacing the Amateur Radio Club's damaged repeater antenna
- Optimized gain and impedance matched antenna with coaxial cable to recover full repeater coverage
- Coordinated material costs as the club's treasurer and recorded purchases for reimbursement

ACTIVITIES, LEADERSHIP & CERTIFICATIONS

- IEEE — Member
- Amateur Radio Club — Treasurer
- FCC Amateur Radio License — General Class (KD3DKR)